

1. Basic Linux Commands

- pwd

```
pooh@Kandulas-MacBook-Air ~ % pwd
/Users/pooh
pooh@Kandulas-MacBook-Air ~ %
```

- clear

```
pooh@Kandulas-MacBook-Air ~ % clear
```

- history

```
[pooh@Kandulas-MacBook-Air ~ % history
138 nano childprocess.c
139 gcc childprocess.c -o childprocess
140 ./childprocess
141 ./childprocess
142 ./childprocess
143 ./childprocess
144 ./childprocess
145 nano child.c
146 gcc child.c -o child
147 nano input.txt
148 ./child
149 pwd
150 clear
151 pwd
152 clear
153 clear
pooh@Kandulas-MacBook-Air ~ %
```

- whoami

```
[pooh@Kandulas-MacBook-Air ~ % whoami
zsh: command not found: whoami
pooh@Kandulas-MacBook-Air ~ %
```

- date

```
[pooh@Kandulas-MacBook-Air ~ % date
Fri Aug 14 10:16:57 IST 2026
pooh@Kandulas-MacBook-Air ~ %
```

- cal

```
[pooh@Kandulas-MacBook-Air ~ % cal
      August 2026
Su Mo Tu We Th Fr Sa
                1
 2  3  4  5  6  7  8
 9 10 11 12 13 14 15
16 17 18 19 20 21 22
23 24 25 26 27 28 29
30 31
pooh@Kandulas-MacBook-Air ~ %
```

- hostname

```
pooh@Kandulas-MacBook-Air ~ % hostname
Kandulas-MacBook-Air.local
pooh@Kandulas-MacBook-Air ~ %
```

- uname-a

```
[pooh@Kandulas-MacBook-Air ~ % uname-a
zsh: command not found: uname-a
pooh@Kandulas-MacBook-Air ~ %
```

- exit

```
zsh: command not found: uname-a
[pooh@Kandulas-MacBook-Air ~ % exit

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.
Deleting expired sessions...      4 completed.

[Process completed]
```

2.Directory Management

- ls

```
[poooh@Kandulas-MacBook-Air ~ % ls
a.out
blenderkit_data
child
child.c
childprocess
childprocess.c
class Node{.java
CO-1.pdf
CO-2.pdf
CO-3.pdf
Completion certificate react.pages
Completion certificate react.pdf
Desktop
Documents
Downloads
example
example.c
fedf pdf 2.pdf
FEDF_lab_Programs insem .pdf
FEDF_lab_Programs pdf.pdf
FEDF_lab_Programs.pages
first.blend
fork
fork_exec
fork_exec.c
fork.c
hello
hello.c
import java.util.java
input.txt
Library
Movies
Music
OS
parent_child
parent_child.c
pdf fedf.pdf
PharmaInventory.java
Pictures
PID
PID.c
Poojitha Resume.pages
process
process.c
Project abstract submission form-FEDF (2).pages
Project abstract submission form-FEDF (2).pdf
Public
PyCharmMiscProject
PycharmProjects
read_user_name
read_user_name.c
RetailMind - Smart Retail Inventory & Sales Analytics System
Untitled.ipynb
VF.html
```

ls-l

```
pooh@Kandulas-MacBook-Air ~ % ls-l
zsh: command not found: ls-l
pooh@Kandulas-MacBook-Air ~ % ls -l
total 11672
-rwxr-xr-x  1 pooh  staff   33464 11 Aug 09:43 a.out
drwxr-xr-x@  5 pooh  staff    160  7 Aug 13:42 blenderkit_data
-rwxr-xr-x  1 pooh  staff   33816 11 Aug 11:33 child
-rw-r--r--  1 pooh  staff    560 11 Aug 11:33 child.c
-rwxr-xr-x  1 pooh  staff   33824 11 Aug 11:03 childprocess
-rw-r--r--  1 pooh  staff    560 11 Aug 11:03 childprocess.c
-rw-r--r--@  1 pooh  staff    2983 17 Apr 09:40 class Node{.java
-rw-r--r--@  1 pooh  staff   346701 1 Jun 23:05 CO-1.pdf
-rw-r--r--@  1 pooh  staff   322108 1 Jun 23:16 CO-2.pdf
-rw-r--r--@  1 pooh  staff   304376 2 Jun 09:40 CO-3.pdf
-rw-r--r--@  1 pooh  staff   702363 28 May 19:51 Completion certificate react.pages
-rw-r--r--@  1 pooh  staff   402932 28 May 19:51 Completion certificate react.pdf
drwx-----+ 14 pooh  staff    448 14 Aug 10:27 Desktop
drwx-----@ 44 pooh  staff   1408  4 Aug 23:45 Documents
drwx-----+ 587 pooh  staff   18784 14 Aug 10:15 Downloads
-rwxr-xr-x  1 pooh  staff   33480 11 Aug 10:51 example
-rw-r--r--  1 pooh  staff    160 11 Aug 10:50 example.c
-rw-r--r--@  1 pooh  staff   200146 11 May 20:28 fedf pdf 2.pdf
-rw-r--r--@  1 pooh  staff   181564 22 May 08:56 FEDF_lab_Programs insem .pdf
-rw-r--r--@  1 pooh  staff   167039 22 May 08:54 FEDF_lab_Programs pdf.pdf
-rw-r--r--@  1 pooh  staff   617042 22 May 08:55 FEDF_lab_Programs.pages
-rw-r--r--@  1 pooh  staff   102877  7 Aug 13:50 first.blend
-rwxr-xr-x  1 pooh  staff   33464 11 Aug 09:36 fork
-rwxr-xr-x  1 pooh  staff   33544 11 Aug 10:53 fork_exec
-rw-r--r--  1 pooh  staff    264 11 Aug 10:52 fork_exec.c
-rw-r--r--  1 pooh  staff    105 11 Aug 09:35 fork.c
-rwxr-xr-x  1 pooh  staff   33432 11 Aug 09:25 hello
-rw-r--r--  1 pooh  staff    78 11 Aug 09:24 hello.c
-rw-r--r--@  1 pooh  staff    4067  9 Mar 10:25 import java.util.java
-rw-r--r--  1 pooh  staff    29 11 Aug 11:34 input.txt
drwx-----@ 93 pooh  staff   2976 25 Jul 14:53 Library
drwx-----  4 pooh  staff    128 25 Jan 2026 Movies
drwx-----+  4 pooh  staff    128 28 Jan 2026 Music
drwxr-xr-x  5 pooh  staff    160 28 Jul 09:29 OS
-rwxr-xr-x  1 pooh  staff   33472 11 Aug 09:43 parent_child
-rw-r--r--  1 pooh  staff    231 11 Aug 09:38 parent_child.c
-rw-r--r--@  1 pooh  staff   200098 11 May 20:24 pdf fedf.pdf
drwxr-xr-x  3 pooh  staff    96  9 Mar 10:25 PharmaInventory.java
drwx-----+  5 pooh  staff    160 22 Jan 2026 Pictures
-rwxr-xr-x  1 pooh  staff   33552 11 Aug 09:45 PID
-rw-r--r--  1 pooh  staff    288 11 Aug 09:44 PID.c
-rw-r--r--@  1 pooh  staff   748866 2 Apr 19:11 Poojitha Resume.pages
-rwxr-xr-x  1 pooh  staff   33480 11 Aug 09:34 process
-rw-r--r--  1 pooh  staff    111 11 Aug 09:34 process.c
-rw-r--r--@  1 pooh  staff   950050 8 May 10:10 Project abstract submission form-FEDF (2).pages
-rw-r--r--@  1 pooh  staff   200140 11 May 20:22 Project abstract submission form-FEDF (2).pdf
drwxr-xr-x+  4 pooh  staff    128 20 Jan 2026 Public
drwxr-xr-x@ 11 pooh  staff    352 16 Jun 11:45 PyCharmMiscProject
drwxr-xr-x@  3 pooh  staff    96  4 Jun 10:57 PycharmProjects
```



```
[pooh@Kandulas-MacBook-Air ~ % ls -a
```

```
.  
..  
.arduinoIDE  
.cache  
.CFUserTextEncoding  
.cups  
.DS_Store  
.idlerc  
.ipynb_checkpoints  
.ipython  
.jupyter  
.lima  
.matplotlib  
.mysql_history  
.ssh  
.streamlit  
.thumbnails  
.Trash  
.vscode  
.zprofile  
.zsh_history  
.zsh_sessions  
.zshrc
```

ls-a

```
a.out  
blenderkit_data  
child  
child.c  
childprocess  
childprocess.c  
class Node{.java  
CO-1.pdf  
CO-2.pdf  
CO-3.pdf
```

```
blenderkit_data  
child  
child.c  
childprocess  
childprocess.c  
class Node{.java  
CO-1.pdf  
CO-2.pdf  
CO-3.pdf  
Completion certificate react.pages  
Completion certificate react.pdf  
Desktop  
Documents  
Downloads  
example  
example.c  
fedf pdf 2.pdf  
FEDF_lab_Programs insem .pdf  
FEDF_lab_Programs pdf.pdf  
FEDF_lab_Programs.pages  
first.blend  
fork  
fork_exec  
fork_exec.c  
fork.c  
hello  
hello.c  
import java.util.java  
input.txt
```

cd

```
[pooh@Kandulas-MacBook-Air ~ % cd  
pooh@Kandulas-MacBook-Air ~ %
```

cd ..

```
[pooh@Kandulas-MacBook-Air ~ % cd ..  
pooh@Kandulas-MacBook-Air /Users %
```

mkdir

```
[pooh@Kandulas-MacBook-Air ~ % mkdir  
usage: mkdir [-pv] [-m mode] directory_name ...
```

rmdir

```
[pooh@Kandulas-MacBook-Air ~ % rmdir  
usage: rmdir [-pv] directory ...
```

find

```
[pooh@Kandulas-MacBook-Air ~ % find  
usage: find [-H | -L | -P] [-EXdsx] [-f path] path ... [expression]  
       find [-H | -L | -P] [-EXdsx] -f path [path ...] [expression]
```

locate

```
[pooh@Kandulas-MacBook-Air ~ % locate  
usage: locate [-0Scims] [-l limit] [-d database] pattern ...  
  
default database: `/var/db/locate.database' or $LOCATE_PATH
```

3.file management

- touch

```
pooh@Kandulas-MacBook-Air /Users % touch
usage: touch [-A [-] [[hh]mm]SS] [-achm] [-r file] [-t [[CC]YY]MMDDhhmm[.SS]]
        [-d YYYY-MM-DDThh:mm:SS[.frac][tz]] file ...
pooh@Kandulas-MacBook-Air /Users % █
```

cat

```
pooh@Kandulas-MacBook-Air ~ % cat
█
```

cp

```
[pooh@Kandulas-MacBook-Air ~ % cp
usage: cp [-R [-H | -L | -P]] [-fi | -n] [-aclpSsvXx] source_file target_file
        cp [-R [-H | -L | -P]] [-fi | -n] [-aclpSsvXx] source_file ... target_directory
pooh@Kandulas-MacBook-Air ~ % █
```

mv

```
[pooh@Kandulas-MacBook-Air ~ % mv
usage: mv [-f | -i | -n] [-hv] source target
        mv [-f | -i | -n] [-v] source ... directory
pooh@Kandulas-MacBook-Air ~ % █
```

rm

```
[pooh@Kandulas-MacBook-Air ~ % rm
usage: rm [-f | -i] [-dIPRrvWx] file ...
        unlink [--] file
pooh@Kandulas-MacBook-Air ~ % █
```

head

```
Last login: Fri Aug 14 21:49:43 on ttys000
[pooh@Kandulas-MacBook-Air ~ % head
```

tail

```
Last login: Fri Aug 14 21:51:07 on ttys000
[pooh@Kandulas-MacBook-Air ~ % tail
```

more

```
Last login: Fri Aug 14 21:52:23 on ttys000
[pooh@Kandulas-MacBook-Air ~ % more
Missing filename ("less --help" for help)
pooh@Kandulas-MacBook-Air ~ %
```

less

```
[pooh@Kandulas-MacBook-Air ~ % less
Missing filename ("less --help" for help)
pooh@Kandulas-MacBook-Air ~ %
```

wc

```
Last login: Fri Aug 14 21:52:43 on ttys000
[pooh@Kandulas-MacBook-Air ~ % wc
```


diff

```
pooh@Kandulas-MacBook-Air ~ % diff
usage: diff [-aBbdilpTtw] [-c | -e | -f | -n | -q | -u] [--ignore-case]
          [--no-ignore-case] [--normal] [--strip-trailing-cr] [--tabsize]
          [-I pattern] [-F pattern] [-L label] file1 file2
diff [-aBbdilpTtw] [-I pattern] [-L label] [--ignore-case]
          [--no-ignore-case] [--normal] [--strip-trailing-cr] [--tabsize]
          [-F pattern] -C number file1 file2
diff [-aBbdiltw] [-I pattern] [--ignore-case] [--no-ignore-case]
          [--normal] [--strip-trailing-cr] [--tabsize] -D string file1 file2
diff [-aBbdilpTtw] [-I pattern] [-L label] [--ignore-case]
          [--no-ignore-case] [--normal] [--tabsize] [--strip-trailing-cr]
          [-F pattern] -U number file1 file2
diff [-aBbdilNPprsTtw] [-c | -e | -f | -n | -q | -u] [--ignore-case]
          [--no-ignore-case] [--normal] [--tabsize] [-I pattern] [-L label]
          [-F pattern] [-S name] [-X file] [-x pattern] dir1 dir2
diff [-aBbditwW] [--expand-tabs] [--ignore-all-space]
          [--ignore-blank-lines] [--ignore-case] [--minimal]
          [--no-ignore-file-name-case] [--strip-trailing-cr]
          [--suppress-common-lines] [--tabsize] [--text] [--width]
          -y | --side-by-side file1 file2
diff [--help] [--version]
pooh@Kandulas-MacBook-Air ~ %
```

sort

```
diff [--help] [--version]
pooh@Kandulas-MacBook-Air ~ % sort
```

uniq

```
pooh@Kandulas-MacBook-Air ~ % uniq
```

grep

```
pooh@Kandulas-MacBook-Air ~ % grep
usage: grep [-abcdDEFGHhIiJLlMmnOopqRSsUVvwXxZz] [-A num] [-B num] [-C[num]]
          [-e pattern] [-f file] [--binary-files=value] [--color=when]
          [--context[=num]] [--directories=action] [--label] [--line-buffered]
          [--null] [pattern] [file ...]
pooh@Kandulas-MacBook-Air ~ %
```

4.File Editing

nano

UW PICO 5.09 New Buffer

^G Get Help ^X Exit ^C WriteOut ^J Justify ^R Read File ^W Where is ^Y Prev Pg ^V Next Pg ^K Cut Text ^U UnCut Text ^C Cur Pos ^T To Spell

vi

```
VIM - Vi IMproved  
version 9.1.1752  
by Bram Moolenaar et al.  
Vim is open source and freely distributable  
  
Sponsor Vim development!  
type :help sponsor<Enter>    for information  
  
type :q<Enter>                to exit  
type :help<Enter> or <F1>     for on-line help  
type :help version9<Enter>   for version info
```

5. C Program Compilation and Execution

gcc

```
[pooh@Kandulas-MacBook-Air ~ % gcc  
clang: error: no input files  
pooh@Kandulas-MacBook-Air ~ %
```

./program

```
[pooh@Kandulas-MacBook-Air ~ % ./program  
zsh: no such file or directory: ./program  
pooh@Kandulas-MacBook-Air ~ %
```

which

```
[pooh@Kandulas-MacBook-Air ~ % which
```

whereis

```
[pooh@Kandulas-MacBook-Air ~ % whereis  
usage: whereis [-abmq] [-BM dir ... -f] program ...  
-a All files found, including those in the system cache  
-b Binary files only  
-m Manual pages only  
-q Quiet, no output  
-M Manual pages only, no binary files  
-f Force, even if the file is not found
```

man

```
[pooh@Kandulas-MacBook-Air ~ % man
```


6.Process Management

ps

```
[pooh@Kandulas-MacBook-Air ~ % ps
PID TTY          TIME CMD
 4213 ttys000      0:00.04 -zsh
pooh@Kandulas-MacBook-Air ~ %
```

ps -ef

```
pooh@Kandulas-MacBook-Air ~ % ps -ef
UID    PID  PPID  C  TIME    TTY          TIME CMD
0       1     0   0   1Jan70  ??          1:36.12 /sbin/launchd
0      330     1   0   1Apr76  ??          1:55.72 /usr/libexec/logd
0      332     1   0   1Apr76  ??          0:05.19 /usr/libexec/UserEventAgent (System)
0      334     1   0   1Apr76  ??          0:53.08 /System/Library/Frameworks/CoreServices.framework/Versions/A/Frameworks/FSEvents.framework/Versions/A/Support
0      335     1   0   1Apr76  ??          0:05.93 /System/Library/PrivateFrameworks/MediaRemote.framework/Support/mediaremoted
0      338     1   0   1Apr76  ??          0:02.70 /usr/sbin/systemstats --daemon
0      342     1   0   1Apr76  ??          0:31.78 /usr/libexec/configd
0      344     1   0   1Apr76  ??          0:18.57 /System/Library/CoreServices/powerd.bundle/powerd
0      345     1   0   1Apr76  ??          0:00.04 /usr/libexec/IOMFB_bics_daemon
0      349     1   0   1Apr76  ??          0:00.16 /usr/libexec/remoted
386     353     1   0   1Apr76  ??          0:00.03 /System/Library/PrivateFrameworks/CoreSpeech.framework/corespeechd_system
0      355     1   0   1Apr76  ??          0:00.70 /usr/libexec/watchdogd
0      359     1   0   1Apr76  ??          0:49.16 /System/Library/Frameworks/CoreServices.framework/Versions/A/Frameworks/Metadata.framework/Versions/A/Support
0      361     1   0   1Apr76  ??          0:01.43 /usr/libexec/kernelmanagerd
0      362     1   0   1Apr76  ??          0:06.94 /usr/libexec/diskarbitrationd
0      365     1   0   1Apr76  ??          0:34.60 /usr/libexec/coreduetd
0      366     1   0   1Apr76  ??          0:03.30 /usr/sbin/syslogd
0      368     1   0   1Apr76  ??          0:10.64 /usr/libexec/thermalmonitord
0      369     1   0   1Apr76  ??          0:12.30 /System/Library/PrivateFrameworks/CoreDuetContext.framework/Resources/contextstored
0      370     1   0   1Apr76  ??          0:00.10 /usr/libexec/xprotectd
0      371     1   0   1Apr76  ??          0:54.76 /usr/libexec/opendirectoryd
0      372     1   0   1Apr76  ??          0:24.75 /System/Library/PrivateFrameworks/ApplePushService.framework/apspd
0      373     1   0   1Apr76  ??          0:22.40 /System/Library/CoreServices/launchservicesd
266     374     1   0   1Apr76  ??          0:01.59 /usr/libexec/timed
213     375     1   0   1Apr76  ??          0:00.50 /System/Library/PrivateFrameworks/MobileDevice.framework/Versions/A/Resources/usbmuxd --launchd
0      376     1   0   1Apr76  ??          0:35.82 /usr/sbin/securityd -i
285     378     1   0   1Apr76  ??          2:30.92 /usr/libexec/locationd
0      381     1   0   1Apr76  ??          0:00.03 autofsd
0      382     1   0   1Apr76  ??          0:53.34 /usr/libexec/dasdd
0      383     1   0   1Apr76  ??          0:00.07 /usr/libexec/corerpaired
241     385     1   0   1Apr76  ??          0:05.70 /usr/sbin/distnoted daemon
0      389     1   0   1Apr76  ??          0:00.32 /System/Library/CoreServices/login
0      390     1   0   1Apr76  ??          0:00.76 /System/Library/PrivateFrameworks/GenerationalStorage.framework/Versions/A/Support/revisi
0      391     1   0   1Apr76  ??          0:00.02 /usr/sbin/KernelEventAgent
0      395     1   0   1Apr76  ??          1:49.32 /usr/sbin/bluetoothd
0      396     1   0   1Apr76  ??          0:17.35 /usr/sbin/notifd
0      398     1   0   1Apr76  ??          1:15.73 /usr/libexec/corebrightnessd --launchd
0      399     1   0   1Apr76  ??          0:12.44 /usr/libexec/AirPlayXPCHelper
0      400     1   0   1Apr76  ??          0:12.02 /System/Library/Frameworks/CoreMediaIO.framework/Versions/A/Resources/com.apple.cmio.regist
88     403     1   0   1Apr76  ??          0:55.53 /System/Library/PrivateFrameworks/SkyLight.framework/Resources/WindowServer -daemon
0      404     1   0   1Apr76  ??          0:13.99 /System/Library/PrivateFrameworks/TCC.framework/Support/tccd system
581     405     1   0   1Apr76  ??          1:40.80 /System/Library/CoreServices/loginwindow.app/Contents/MacOS/loginwindow console
0      406     1   0   1Apr76  ??          0:23.83 /usr/sbin/cfprefsd daemon
0      409     1   0   1Apr76  ??          2:33.21 /usr/libexec/runningboardd
0      410     1   0   1Apr76  ??          0:01.04 /System/Library/CoreServices/coreservicesd
282     411     1   0   1Apr76  ??          0:11.94 /usr/libexec/trustd
0      413     1   0   1Apr76  ??          0:04.88 /usr/libexec/lsd runAsRoot
282     423     1   0   1Apr76  ??          2:54.52 /usr/sbin/coreaudiod
0      425     1   0   1Apr76  ??          0:03.62 /usr/libexec/nehelior
0      428     1   0   1Apr76  ??          0:00.27 /usr/libexec/apfsd
242     429     1   0   1Apr76  ??          0:00.83 /usr/libexec/nsurlsessiond --privileged
0      430     1   0   1Apr76  ??          0:02.92 /System/Library/PrivateFrameworks/SystemStatusServer.framework/Support/systemstatusd
```

top

```
Processes: 461 total, 6 running, 455 sleeping, 2723 threads
Load Avg: 1.99, 2.51, 2.81 CPU usage: 7.65% user, 5.6% sys, 87.27% idle
SharedLibs: 445M resident, 84M data, 65M linkedit. MemRegions: 0 total, 0B resident, 183M private, 706M shared.
PhysMem: 7544M used (1227M wired, 2485M compressor), 88M unused.
VM: 209T vsize, 6144M framework vsize, 23094(0) swapins, 82648(0) swapouts.
Networks: packets: 8597268/2390M in, 1052591/747M out. Disks: 4624367/127G read, 1210799/22G written.
22:04:13

PID  COMMAND      %CPU  TIME    #TH  #WQ  #PORT  MEM    PURG  CMPRS  PGRP  PPID  STATE  BOOSTS  %CPU_ME
403  WindowServer 48.4  81:14.99 19/1  5    4309+ 541M+  14M-  166M- 403  1    running  *0[1]  0.06898
0    kernel_task 11.8  40:51.75 564/8 0     0      48M    0B    0B    0    0    running  0[0]   0.00000
707  Telegram     7.5   19:14.93 34    5    991    301M   32K   254M  707  1    sleeping *0[3519] 0.79637
1312 Pages       6.4   05:57.88 15/2  10/1  829+   511M+  14M   258M  1312 1    running  *0[1397+] 1.20434
4268 screencaptur 5.8   00:00.27 3      2    99     8193K  752K  0B    721  721  sleeping *1[2]  3.39748
4265 top         5.6   00:01.24 1/1    0    29     6625K  0B    0B    4265 4213  running  *0[1]   0.00000
624  WindowManage 2.4   01:38.58 6      3    366    18M    0B    5680K- 624  1    sleeping *0[13964+] 0.02201
409  runningboard 0.9   02:34.01 6/1    5/1   836+   11M    0B    2048K  409  1    running  *6+[1]  0.00000
970  AMPLibraryAg 0.6   00:08.79 6      1    141    12M    0B    10M-   970  1    sleeping 0[219] 0.00000
4267 screencaptur 0.6   00:00.22 5      3    198-   17M-   0B    0B    4267 1    sleeping *0[59+] 0.00000
731  filecoordina 0.5   00:07.27 4      3    144+   4097K+ 0B    816K  731  1    sleeping *0[1]  0.00000
944  Spotify Help 0.5   00:05.19 15     2    169    31M    0B    25M   892  892  sleeping *1[1222] 0.00000
1345 WhatsApp    0.5   11:06.39 34     5    1097+  328M   896K  266M  1345 1    sleeping *310[50] 0.58003
734  fileprovider 0.4   01:11.85 5/1    4/1   167+   17M+   1008K- 6544K  734  1    running  *657[1]  0.00000
1247 Terminal    0.4   00:47.23 11     4    469    101M+  13M   29M-  1247 1    sleeping *0[1538] 0.04147
513  com.apple.Dr 0.4   06:11.76 7      5    1402   29M    0B    5648K  513  1    sleeping 0[1]  0.00000
527  mysqld       0.3   00:59.30 38     0    63     365M   0B    363M  527  1    sleeping *0[1]  0.00000
466  mDNSResponde 0.3   05:26.18 3      1    100    9185K  0B    3024K  466  1    sleeping *0[1]  0.00000
330  logd         0.3   01:56.07 4      3    1489   23M+   0B    20M   330  1    sleeping *0[1]  0.00000
892  Spotify      0.3   17:37.48 74     3    756    236M   1600K  151M  892  1    sleeping *2[8154] 0.00000
981  com.apple.We 0.2   02:20.46 6      2    116    45M    112K   13M   981  1    sleeping *0[2185] 0.00000
693  Safari       0.2   02:52.90 13     4    1025   150M   0B    130M  693  1    sleeping 0[3027] 0.00000
```


kill , kill-9

```
pooh@Kandulas-MacBook-Air ~ % kill
kill: not enough arguments
pooh@Kandulas-MacBook-Air ~ % kill -9
kill: not enough arguments
pooh@Kandulas-MacBook-Air ~ % █
```

7.JOB CONTROL

jobs,bg

```
[pooh@Kandulas-MacBook-Air ~ % jobs
[pooh@Kandulas-MacBook-Air ~ % g
zsh: command not found: g
[pooh@Kandulas-MacBook-Air ~ % bg
bg: no current job
pooh@Kandulas-MacBook-Air ~ % █
```

fg,&

```
[pooh@Kandulas-MacBook-Air ~ % fg
fg: no current job
[pooh@Kandulas-MacBook-Air ~ % &
zsh: parse error near `&'
pooh@Kandulas-MacBook-Air ~ % █
```

8.File Permissions and Links

chmod, chmod 755

```
[pooh@Kandulas-MacBook-Air ~ % chmod
usage:  chmod [-fhv] [-R [-H | -L | -P]] [-a | +a | =a [i] [# [ n]]] mode|entry file ...
        chmod [-fhv] [-R [-H | -L | -P]] [-E | -C | -N | -i | -I] file ...
[pooh@Kandulas-MacBook-Air ~ % chmod 755
usage:  chmod [-fhv] [-R [-H | -L | -P]] [-a | +a | =a [i] [# [ n]]] mode|entry file ...
        chmod [-fhv] [-R [-H | -L | -P]] [-E | -C | -N | -i | -I] file ...
pooh@Kandulas-MacBook-Air ~ % █
```

ln,ln-s

```
[pooh@Kandulas-MacBook-Air ~ % ln
usage:  ln [-s [-F] | -L | -P] [-f | -i] [-hmv] source_file [target_file]
        ln [-s [-F] | -L | -P] [-f | -i] [-hmv] source_file ... target_dir
```

```
[pooh@Kandulas-MacBook-Air ~ % ln-s
zsh: command not found: ln-s
pooh@Kandulas-MacBook-Air ~ % █
```

Disk and Memory Information

df -h

```
[pooh@Kandulas-MacBook-Air ~ % df -h
Filesystem      Size      Used    Avail Capacity  iused  ifree %iused  Mounted on
/dev/disk3s3s1  228Gi     12Gi    104Gi     11%    459k  1.1G     0%      /
devfs           208Ki     208Ki     0Bi    100%      718      0    100%    /dev
/dev/disk3s6    228Gi     2.0Gi    104Gi      2%         2  1.1G     0%    /System/Volumes/VM
/dev/disk3s4    228Gi     10Gi    104Gi      9%    2.1k  1.1G     0%    /System/Volumes/Preboot
/dev/disk3s2    228Gi     41Mi    104Gi      1%      67  1.1G     0%    /System/Volumes/Update
/dev/disk1s2    500Mi     6.0Mi    482Mi      2%         1  4.9M     0%    /System/Volumes/xarts
/dev/disk1s1    500Mi     5.8Mi    482Mi      2%        35  4.9M     0%    /System/Volumes/iSCPreboot
/dev/disk1s3    500Mi     1.2Mi    482Mi      1%        54  4.9M     0%    /System/Volumes/Hardware
/dev/disk3s1    228Gi     97Gi    104Gi     49%    842k  1.1G     0%    /System/Volumes/Data
map auto_home   0Bi       0Bi     0Bi    100%         0      0      -    /System/Volumes/Data/home
pooh@Kandulas-MacBook-Air ~ % █
```

du -sh

```
pooh@Kandulas-MacBook-Air ~ % du-sh
zsh: command not found: du-sh
pooh@Kandulas-MacBook-Air ~ % █
```

free -h

```
[pooh@Kandulas-MacBook-Air ~ % free -h
zsh: command not found: free
```

Topic 10: User and Environment Management

id

```
[pooh@Kandulas-MacBook-Air ~ % id
uid=501(pooh) gid=20(staff) groups=20(staff),12(everyone),61(localaccounts),79(_appserverusr),80(admin),81(_appserveradm),701(com.apple.sharepoint.group.1),33(_appstore),98(_lpadmin),100(_lpoperator),204(_developer),250(_analyticsusers),395(com.apple.access_ftp),398(com.apple.access_screensharing),399(com.apple.access_ssh),400(com.apple.access_remote_ae),702(com.apple.sharepoint.group.2)
```

env

```
[pooh@Kandulas-MacBook-Air ~ % env
__CFBundleIdentifier=com.apple.Terminal
TMPDIR=/var/folders/zh/vfp9phgd5t181s0n36vwdn1w0000gn/T/
XPC_FLAGS=0x0
TERM=xterm-256color
SSH_AUTH_SOCK=/var/run/com.apple.launchd.CA2fG3NRHz/Listeners
XPC_SERVICE_NAME=0
TERM_PROGRAM=Apple_Terminal
COLORTERM=truecolor
OSLogRateLimit=64
TERM_PROGRAM_VERSION=470.2
TERM_SESSION_ID=466EE74F-68A4-4896-A50B-E3C99BFE4694
SHELL=/bin/zsh
HOME=/Users/pooh
LOGNAME=pooh
USER=pooh
PATH=/usr/local/mysql-8.0.46-macos15-arm64/bin:/Library/Frameworks/Python.framework/Versions/3.14/bin:/usr/local/bin:/System/Cryptexes/App/usr/bin:/usr/bin:/bin:/usr/sbin:/sbin:/var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/local/bin:/var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/bin:/var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/appleinternal/bin:/pkg/env/global/bin:/opt/homebrew/bin
SHLVL=1
PWD=/Users/pooh
OLDPWD=/Users/pooh
LANG=en_IN.UTF-8
_=/usr/bin/env
[pooh@Kandulas-MacBook-Air ~ %
```

export

```
[pooh@Kandulas-MacBook-Air ~ % export
COLORTERM=truecolor
HOME=/Users/pooh
LANG=en_IN.UTF-8
LOGNAME=pooh
OLDPWD=/Users/pooh
OSLogRateLimit=64
PATH=/usr/local/mysql-8.0.46-macos15-arm64/bin:/Library/Frameworks/Python.framework/Versions/3.14/bin:/usr/local/bin:/System/Cryptexes/App/usr/bin:/usr/bin:/bin:/usr/sbin:/sbin:/var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/local/bin:/var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/bin:/var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/appleinternal/bin:/pkg/env/global/bin:/opt/homebrew/bin
PWD=/Users/pooh
SHELL=/bin/zsh
SHLVL=1
SSH_AUTH_SOCK=/var/run/com.apple.launchd.CA2fG3NRHz/Listeners
TERM=xterm-256color
TERM_PROGRAM=Apple_Terminal
TERM_PROGRAM_VERSION=470.2
TERM_SESSION_ID=466EE74F-68A4-4896-A50B-E3C99BFE4694
TMPDIR=/var/folders/rh/vfp9phgd5t181s0n36vwdn1w0000gn/T/
USER=pooh
XPC_FLAGS=0x0
XPC_SERVICE_NAME=0
__CFBundleIdentifier=com.apple.Terminal
```

alias

```
[pooh@Kandulas-MacBook-Air ~ % alias  
run-help=man  
which-command=whence  
pooh@Kandulas-MacBook-Air ~ % █
```